

# SEC P vs NP — notebook summary

SEC (autonomous) — attributed to Ludovico Kubler

2026-04-28 19:42 UTC

## SEC P vs NP — notebook summary

Generated 2026-04-28 19:42 UTC

- Cycles recorded: **205**
- Time span: 118.0h (~1.74 cycles/h)
- Notebook: /home/ludo/Scrivania/SEC/research/pvsnp\_notebook.jsonl

### Verdict distribution

| Verdict      | Count |
|--------------|-------|
| INCONCLUSIVE | 182   |
| FALSIFIED    | 14    |
| BARRIER_HIT  | 5     |
| SUPPORTED    | 4     |

### Mathematical fields explored (field\_A)

| Field                                                                             | Cycles |
|-----------------------------------------------------------------------------------|--------|
| Ergodic Circuit Framework (communication complexity via dynamical systems)        | 4      |
| {“framework_name”: “Ergodic Circuit Framework”, “math_branch”: “COMM_COMPLEXITY”} | 4      |
| Fourier analysis of boolean functions                                             | 3      |
| Noncommutative geometry                                                           | 2      |
| Representation theory of symmetric groups                                         | 2      |
| Ergodic Circuit Framework (COMM_COMPLEXITY)                                       | 2      |
| Matroid Theory                                                                    | 2      |
| Algebraic Topology                                                                | 2      |
| Tropical geometry                                                                 | 2      |
| Random Matrix Theory                                                              | 2      |
| Combinatorial homotopy theory                                                     | 1      |
| Kazhdan-Lusztig theory of Hecke algebras                                          | 1      |
| Combinatorial algebraic topology                                                  | 1      |
| Knot theory (quantum invariants)                                                  | 1      |
| Quadratic forms over finite fields (Grothendieck-Witt groups)                     | 1      |
| Clifford algebras over finite fields                                              | 1      |
| Toric geometry (via Gröbner degenerations)                                        | 1      |

| Field                                                              | Cycles |
|--------------------------------------------------------------------|--------|
| Arithmetic geometry (Tate-Shafarevich groups)                      | 1      |
| Geometric Langlands correspondence                                 | 1      |
| Algebraic graph theory (chromatic polynomials)                     | 1      |
| Clifford algebras over real vector spaces                          | 1      |
| Algebraic cycles (Chow groups over finite fields)                  | 1      |
| Algebraic lattice theory (Möbius functions of flow lattices)       | 1      |
| Directed algebraic topology (d-space homology)                     | 1      |
| Motivic integration (Denef-Loeser zeta functions)                  | 1      |
| Matroid theory (Clifford index of binary matroids)                 | 1      |
| Representation theory of finite groups (Frobenius-Schur indicator) | 1      |
| Categorification and knot Floer homology                           | 1      |
| Modular forms                                                      | 1      |
| Algebraic geometry (ideals in polynomial rings)                    | 1      |

### Last 15 cycles (chronological)

| Time                 | Verdict      | Title                                                                             |
|----------------------|--------------|-----------------------------------------------------------------------------------|
| 2026-04-28 12:48 UTC | INCONCLUSIVE | Bipartite Token-Sliding Diameter Lower-Bounds                                     |
| 2026-04-28 13:27 UTC | INCONCLUSIVE | Monotone-KW Depth Magnus Level-2 Defect Bounds DNF_min via Truth-Table Inversions |
| 2026-04-28 13:54 UTC | INCONCLUSIVE | Monomer-Dimer Entropy Lower-Bounds Tseitin Resolution Width                       |
| 2026-04-28 14:25 UTC | INCONCLUSIVE | Sprague-Grundy Game Value of CNF Bounds Tree-Frege Lines                          |
| 2026-04-28 14:45 UTC | INCONCLUSIVE | Roe-Skeleton Spectral Gap Forces Multiplicity in Protocol-Induced                 |
| 2026-04-28 14:54 UTC | INCONCLUSIVE | Walsh-Code Antichain Width of NW Designs Bounds PRG Linear Bias                   |
| 2026-04-28 15:27 UTC | INCONCLUSIVE | Kruskal-Katona Shadow Defect Lower-Bounds Monotone-KW Depth                       |
| 2026-04-28 16:05 UTC | INCONCLUSIVE | Hodge Laplacian $b_1$ of Edge-Triangle Star-Complex                               |
| 2026-04-28 16:32 UTC | INCONCLUSIVE | Bounds SoS Degr Continued-Fraction Spike of Truth-Table Rational Caps             |
| 2026-04-28 16:59 UTC | INCONCLUSIVE | DNF_min Shifted Partialis of Clause Cubic Lower-Bound Tree-Resolution Leav        |

| Time                 | Verdict      | Title                                                                                         |
|----------------------|--------------|-----------------------------------------------------------------------------------------------|
| 2026-04-28 17:25 UTC | INCONCLUSIVE | Hereditary Discrepancy of Clause-Variable Incidence Lower-Bounds                              |
| 2026-04-28 17:55 UTC | INCONCLUSIVE | Schur-Polynomial Positivity Defect Lower-Bounds                                               |
| 2026-04-28 18:31 UTC | INCONCLUSIVE | Tree-Frege Lines Erdős-Rado Sunflower Petals Lower-Bound Lifted Log-Rank for IND <sub>2</sub> |
| 2026-04-28 19:01 UTC | INCONCLUSIVE | Subword Complexity of Lifted Rows Lower-Bounds Real Rank for IND <sub>2</sub>                 |
| 2026-04-28 19:42 UTC | INCONCLUSIVE | Bakry-Émery Curvature Floor Lower-Bounds Tseitin Resolution Width                             |

## How to read the reports

- reports/supported\_findings.md — SUPPORTED conjectures with full test code. Paper-quality.
- reports/falsified\_findings.md — FALSIFIED with counterexamples.
- reports/daily\_YYYY-MM-DD.md — chronological log by day.
- pvsnp\_notebook.jsonl — raw JSONL, one entry per cycle (source of truth).

## PDF export

Install pandoc + a LaTeX engine once (`sudo apt install pandoc texlive-xetex`), then:

```
cd research/reports
for f in supported_findings.md falsified_findings.md notebook_summary.md; do
    pandoc "$f" -o "${f%.md}.pdf" --pdf-engine=xelatex -V geometry:margin=2cm
done
```